Presentation and challenge session

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Show the aggregate result, failure evidence and honest unknowns rather than only a best demonstration.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Final project manuscript. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Figures that preserve **evidence**. Give an example and a mistake that the idea helps you avoid.

Final project manuscript

> **My remembered idea, example and correction**

Peer review and response

> **My remembered idea, example and correction**

Figures that preserve evidence

> **My remembered idea, example and correction**

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A research presentation makes the evidence challengeable: question, design, primary result, failure, limitation and next question. Main one perfect output does not replace the fourteen/twenty study rate of 70%. Other six cases are not supplied as passes; exact failure/unassessed detail should come from actual records, not guesses.

  Separate original camera-A study has ten of twelve correct, about 83.333%, with no camera-B records. Answer the B challenge by stating that boundary and proposing reserved labelled B cases, grouping and fixed scoring. A later one-off B success is a demonstration, not broad condition coverage or a representative reliability estimate. Keep original and later records separate.

  Proposed work stays not run until evidence exists. Sampling and uncertainty remain relevant. A fictional desktop-only project similarly cannot establish mobile performance. The course performs a paper presentation rehearsal, not a live audience interaction or new study.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  Proposed work stays not run until evidence exists. Sampling and uncertainty remain relevant. A fictional desktop-only project similarly cannot establish mobile performance. The course performs a paper presentation rehearsal, not a live audience interaction or new study.

- A presenter shows one perfect **output** after a 20-case study with 14 passes. Show the intermediate steps and give the conclusion with its scope.

Main presentation step	Evidence-grounded version
Selected perfect output	Identify as one demonstration
Aggregate primary result	Fourteen of twenty, or seventy percent
Remaining six cases	Show actual failure/unassessed record when available; do not invent breakdown
Claim best output is representative	Not established by selecting it
Failure/limitation slide	Retain concrete supplied evidence and gaps
Next question	Propose a bounded test, with not-run status

Main presentation step	Evidence-grounded version
Audience question outside scope	Locate evidence or state it is absent

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A research presentation makes the evidence challengeable: question, design, primary result, failure, limitation and next question. Main one perfect output does not replace the fourteen/twenty study rate of 70%. Other six cases are not supplied as passes; exact failure/unassessed detail should come from actual records, not guesses.

  Separate original camera-A study has ten of twelve correct, about 83.333%, with no camera-B records. Answer the B challenge by stating that boundary and proposing reserved labelled B cases, grouping and fixed scoring. A later one-off B success is a demonstration, not broad condition coverage or a representative reliability estimate. Keep original and later records separate.

  Proposed work stays not run until evidence exists. Sampling and uncertainty remain relevant. A fictional desktop-only project similarly cannot establish mobile performance. The course performs a paper presentation rehearsal, not a live audience interaction or new study.

- A report omits this qualification: A confident response should not replace missing **evidence** with speculation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A research presentation makes the evidence challengeable: question, design, primary result, failure, limitation and next question. Main one perfect output does not replace the fourteen/twenty study rate of 70%. Other six cases are not supplied as passes; exact failure/unassessed detail should come from actual records, not guesses.

  Separate original camera-A study has ten of twelve correct, about 83.333%, with no camera-B records. Answer the B challenge by stating that boundary and proposing reserved labelled B cases, grouping and fixed scoring. A later one-off B success is a demonstration, not broad condition coverage or a representative reliability estimate. Keep original and later records separate.

  Proposed work stays not run until evidence exists. Sampling and uncertainty remain relevant. A fictional desktop-only project similarly cannot establish mobile performance. The course performs a paper presentation rehearsal, not a live audience interaction or new study.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

  Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Presentation and **challenge session**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Presentation and **challenge session**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Audience asks if result holds on mobile; only desktop was tested. Give an honest answer.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.
- The project tested twelve photos from camera A only. Ten were correct; no camera B records exist.
- Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Answer the challenge from the evidence you have

Camera A: tested

10 / 12 correct

≈83.333% on supplied photos

Camera B: original study

NO RECORDS

Do not promise an untested rate

Later one camera-B success: one demonstration, not broad reliability.

Propose reserved camera-B cases, fixed scoring and recorded grouping.
Mark the proposed test as not run; keep original and demonstration records separate.

AI ACADEMY / AI-7/35

Presentation and challenge session

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. State measured result

Report 10/12 correct on camera A.

2. Address the challenge

Say camera B performance is not measured.

3. Propose a valid next test

Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

CHANGE ONE THING

The reviewer shows one camera-B success.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera.
- Which labelled step matches this detail? Use reserved camera-B cases with labels, grouping and a fixed scoring rule.
- 1. Propose a valid next test
- 2. State measured result
- 3. Address the challenge
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Propose a valid next test Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

Changed-case question and optional hint

- Changed-case practice: The reviewer shows one camera-B success. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. That is one demonstration, not an estimate of broad camera-B reliability.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That is one demonstration, not an estimate of broad camera-B reliability. One selected **observation** does not represent the full proposed condition.

Repair and retain the record

Review Final project manuscript (ai-7/34). Goal: Explain the **mechanism** and terminology of final project manuscript. Short practice: Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping.

Which abstract claim must be corrected if it says ten independent plants?. Return to this week and explain what you changed.

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Figures that preserve **evidence** (ai-7/29). Goal: Explain the **mechanism** and terminology of figures that preserve **evidence**. Short practice: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Presentation and challenge session: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

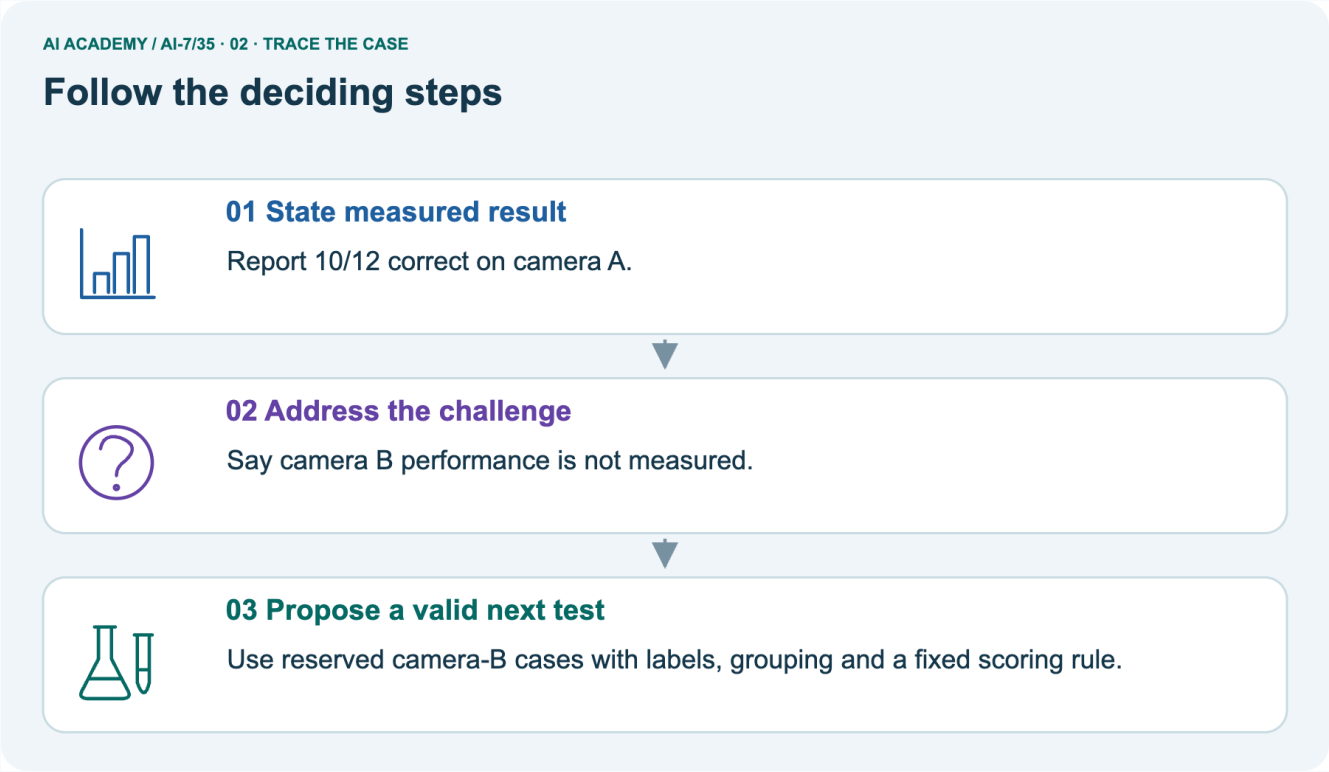
Answer a challenge without inventing **evidence**. At the presentation, a reviewer asks whether the model works on a different camera. Supplied facts: The project tested twelve photos from camera A only.

Ten were correct; no camera B records exist.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



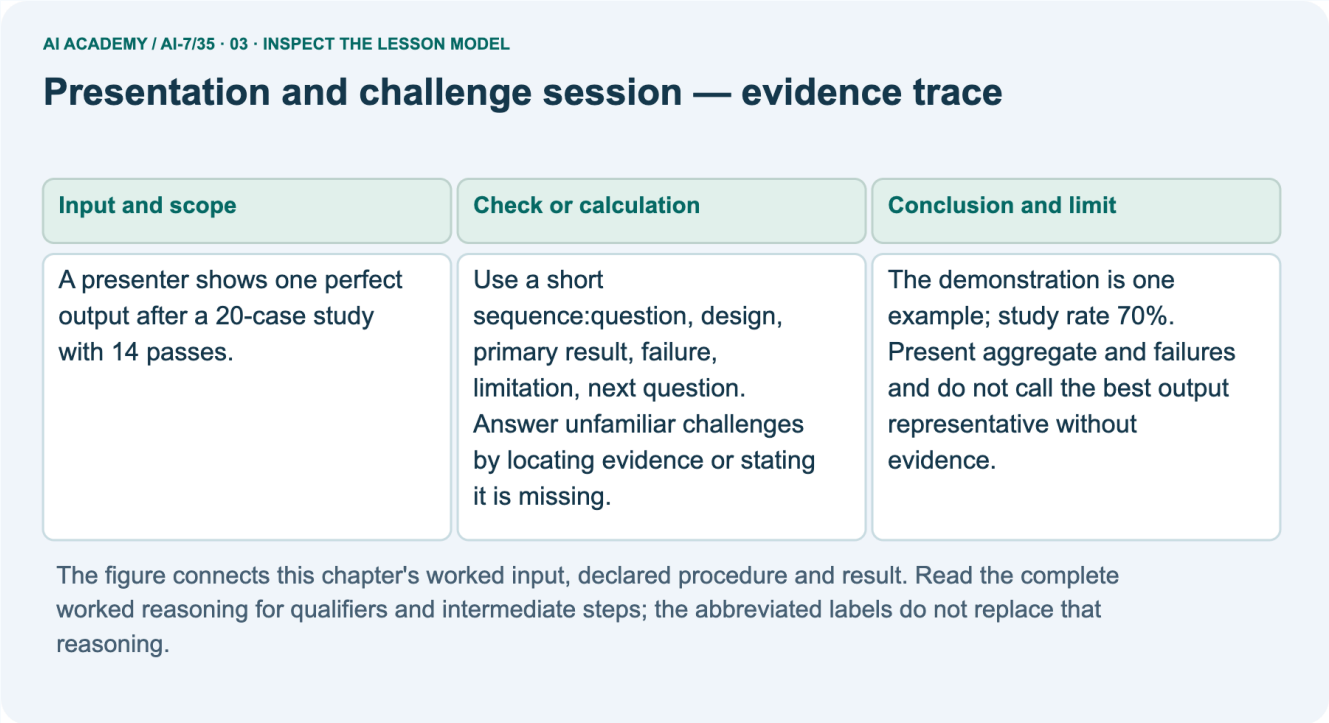
Follow the labelled picture with the matching explanation. Open the vector diagram

State measured result: Report 10/12 correct on camera A. → Address the challenge: Say camera B performance is not measured. → Propose a valid next test: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Presentation and challenge session — evidence trace



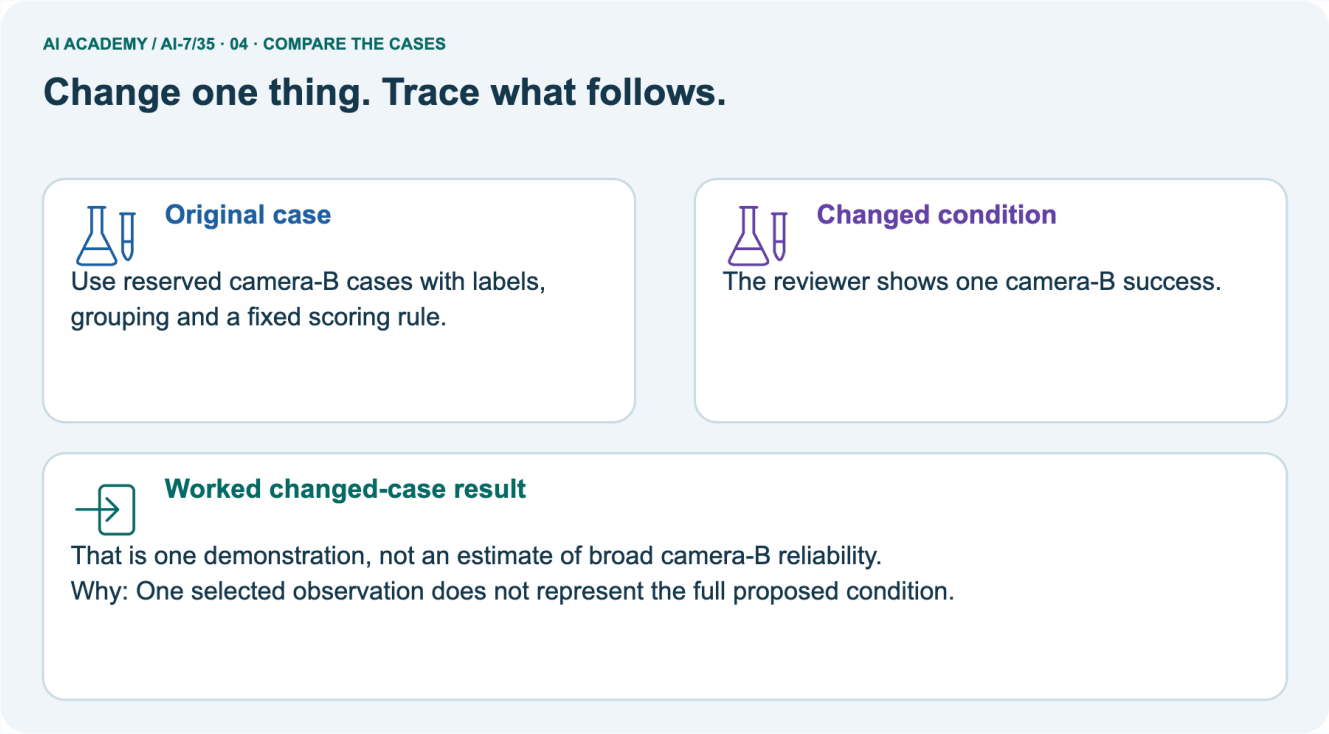
Follow the labelled picture with the matching explanation. Open the vector diagram

Presentation and **challenge session** — **evidence** trace. A presenter shows one perfect **output** after a 20-case study with 14 passes. | Use a short sequence:question, design, primary result, failure, limitation, next question. Answer unfamiliar challenges by locating **evidence** or stating it is missing. | The demonstration is one example; study rate **70%**. Present aggregate and failures and do not call the best **output** representative without **evidence**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

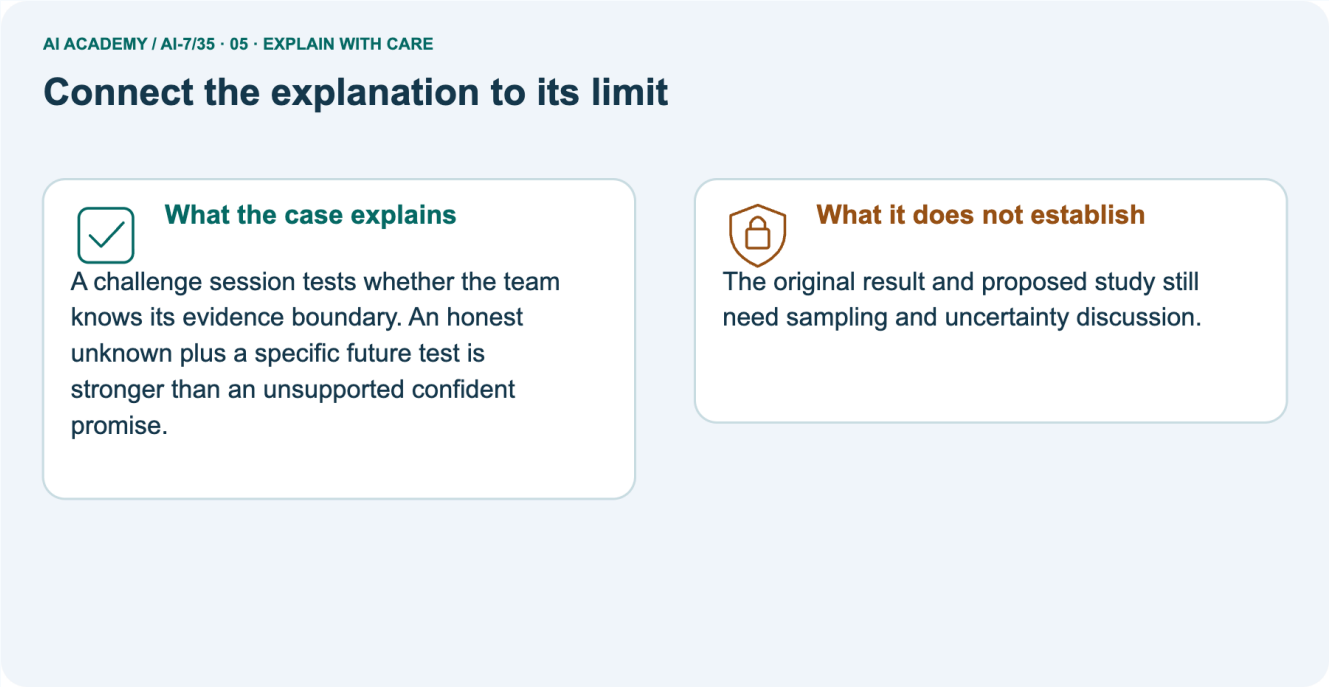
Original case: Use reserved camera-B cases with labels, grouping and a fixed scoring rule.
Changed condition: The reviewer shows one camera-B success.

Expected result: That is one demonstration, not an estimate of broad camera-B reliability.
Reason: One selected **observation** does not represent the full proposed condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

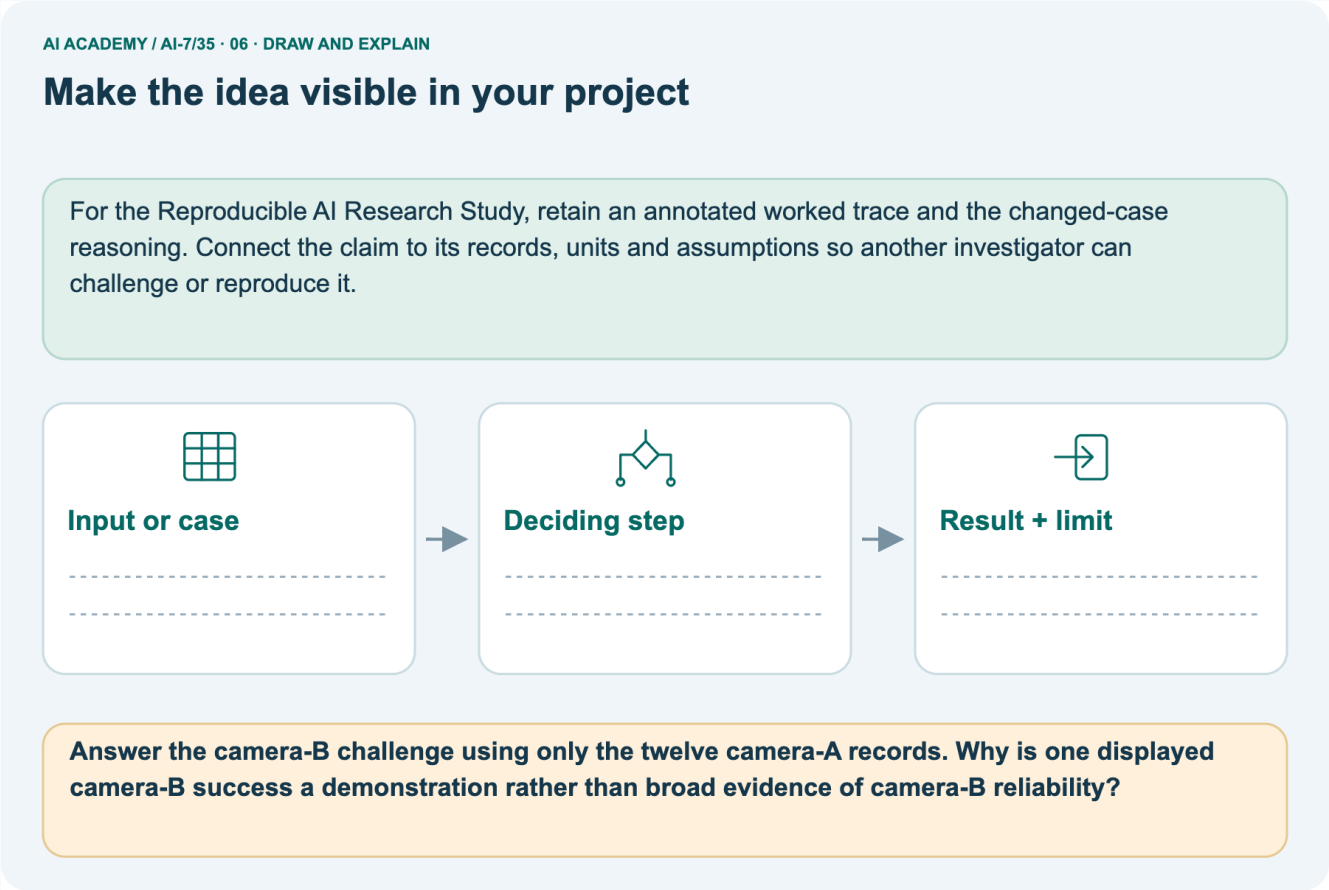
A **challenge session** tests whether the team knows its **evidence** boundary. An honest unknown plus a specific future test is stronger than an unsupported confident promise.

Limit: The original result and proposed study still need sampling and uncertainty discussion.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Answer the camera-B challenge using only the twelve camera-A records.

Why is one displayed camera-B success a demonstration rather than broad **evidence** of camera-B reliability?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Presentation and challenge session: the case

02. Follow the deciding steps

03. Presentation and challenge session — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.